

Question Paper

Exam Date & Time: 29-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
Digital System Design and Computer Organization [ECM_2122]

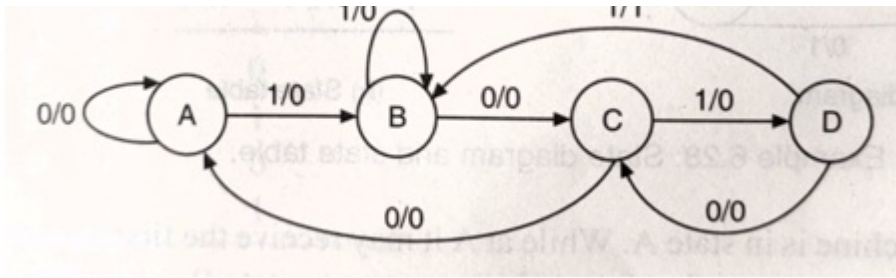
Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Implement the following Boolean functions using PAL with four inputs and 3-wide AND-OR structure (4)
 $F1(A, B, C, D) = \sum m(7, 8, 9, 10, 11, 12, 13, 14, 15)$
A) $F2(A, B, C, D) = \sum m(1, 2, 8, 12, 13)$
Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Implement a mod-12 ripple up-counter using negative edge triggered JK flipflops and draw the output timing diagrams. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Write a Verilog structural code to implement 4:1 multiplexer using instances of 2:1 multiplexers in dataflow. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 2) Design a 3-bit self-correcting synchronous counter using JK flip-flops that follows the sequence 0-1-2-4 repeatedly, ensuring that any unused states automatically go to the 0-state on the next clock pulse. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) What is the role of Cache in computer memory organization? If a cache has 95% hit ratio, an access time of 100 ns on a cache-hit, and an access time of 800 ns on a cache-miss, compute the effective access time. (3)
Course Outcome: "CO5" / Cognitive Level: "Understand"
- C) Differentiate between hardware description language (such as Verilog) and C language. (2)
Course Outcome: "CO3" / Cognitive Level: "Understand"
- 3) Design a Mealy synchronous sequential circuit using D flip-flops that will function as prescribed by the state diagram shown below. (5)
A)



Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Illustrate the use of Booth's algorithm to multiply -11 (multiplicand) and +8 (multiplier), where each number is represented using 5 bits. (3)

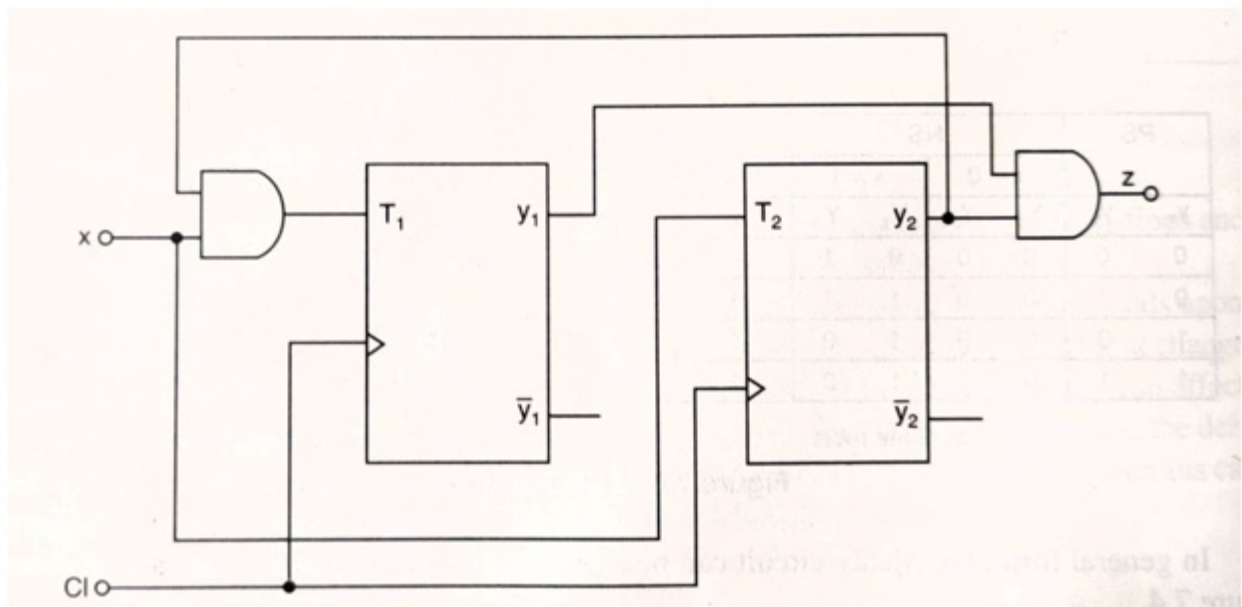
Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Write a switch-level Verilog code for CMOS inverter using NMOS and PMOS primitives. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Analyze the synchronous sequential circuit shown below and obtain its state table and state diagram. (4)

A)



Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Express the given statement in three, two and one instruction addressing formats: $D = (A+B) \times (C+D)$ (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) What are the different hazards in pipelining? Explain them with an example for each. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5) Write the Verilog code in behavioral modeling to implement a 3-bit synchronous binary counter using negative edge-triggered T flip-flops. (4)

A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Define the following addressing modes with an example for each type: (i) register-indirect, (ii) autoincrement, (iii) indexed (displacement) and (iv) immediate (4)

- C) Explain the operation of a general register-based computer organization. (2)

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECM_2122:Digital System Design and Computer Organization

#	Course Outcome	Description	Marks
1	1	CO 1	4
2	2	CO 2	17
3	3	CO 3	11
4	4	CO 4	10
5	5	CO 5	8
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECM_2122:Digital System Design and Computer Organization

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	13(26%)
3	Application	10	37(74%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)